

Esolution

Place student sticker here

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Statistics for Business Administration

Exam: MA9712 / Final
Examiner: Prof. Mathias Drton

Date: Saturday 29th July, 2023
Time: 10:30 – 12:00

Working instructions

- This exam consists of **14 pages** with a total of **8 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 42 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **scientific pocket calculator**
 - the **cheat sheet**, which is contained in this exam on page **13**
- Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Additional space for solution to **any** of the problems is given on page 8, 12 and 14.
- Do not write with red or green colors nor use pencils.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

Give a short answer to each of the following sub-problems.

Give a short answer to each of the following sub-problems.

-
- A blank sheet of graph paper with a grid pattern. The grid consists of small squares, with a larger square area at the bottom right corner. There are some faint markings on the left side, possibly from a previous page or binding.

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- A blank sheet of graph paper with a grid pattern. The grid consists of small squares formed by thin black lines. There are 20 columns and 10 rows of squares. A thicker vertical line runs down the left side, creating a margin. A thicker horizontal line runs across the top, creating a header area. The paper is otherwise empty of any markings or text.

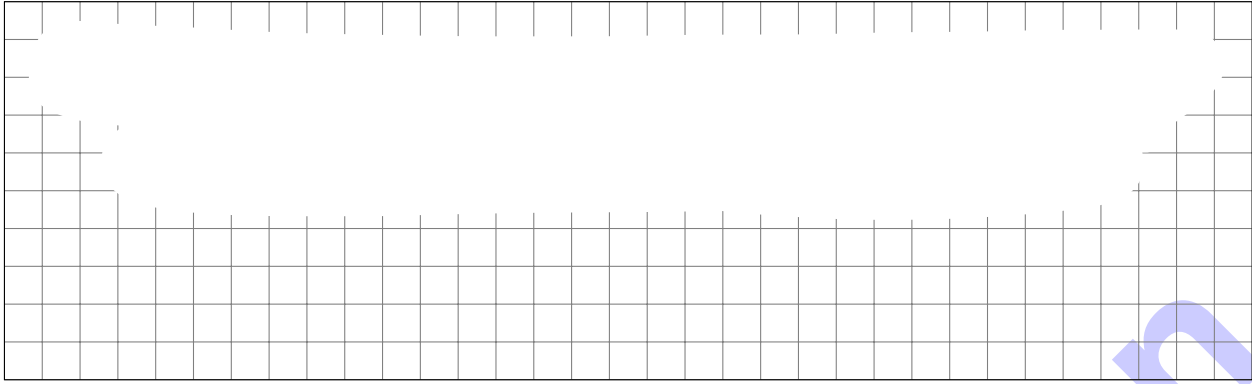
- [illegible]

- | Grades | 1 | 2 | 3 | 4 | 5 |
|--------|-----|------|-----|-----|------|
| P_a | 0.3 | 0.3 | 0.3 | 0.2 | 0.1 |
| P_b | 0 | 0 | 1 | 0 | 0 |
| P_c | 0.3 | 0.3 | 0.3 | 0 | 0 |
| P_d | 0.3 | 0.5 | 0.2 | 0.1 | -0.1 |
| P_e | 0.2 | 0.4 | 0.2 | 0.1 | 0.1 |
| P_f | 0 | -0.1 | 1.1 | 0 | 0 |

[illegible]

e) Suppose we have n independent random variables $X_1, X_2, \dots, X_n$, such that $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2$ for all $i \in \{1, \dots, n\}$. Calculate the variance of the random variable $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.

0
1
2

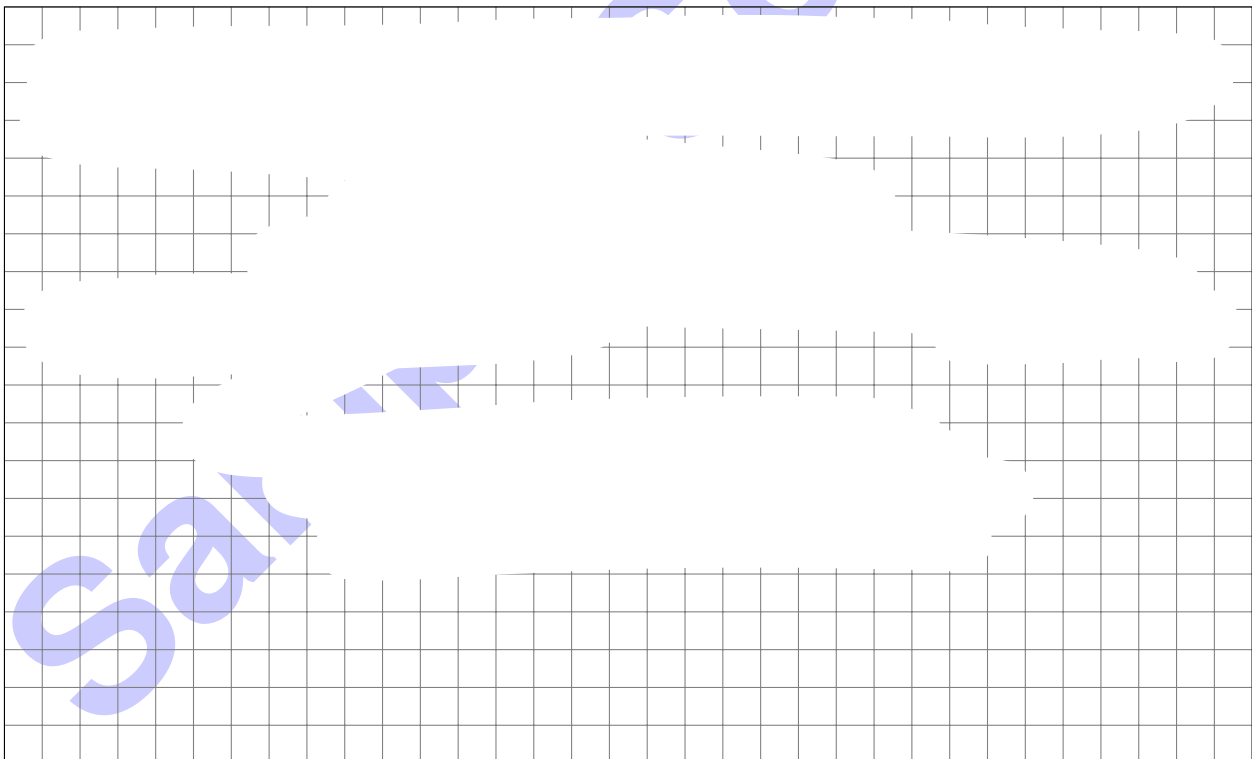


Problem 2 (3 credits)

Jose uses the parking lot at the train station Olympiazentrum every Thursday evening. However, some days the parking lot is full, often due to events in the nearby Olympiapark. There are cultural events on 34% of evenings, sporting events on 28% of evenings, and no events on 38% of evenings. When there is a cultural event, the parking lot fills up about 22% of the time, and it fills up 67% of evenings with sporting events. On evenings when there are no events, it only fills up about 11% of the time.

If Jose comes to the parking lot and finds it full, what is the probability that there is a sporting event? Write down all steps, which are needed to compute the wanted probability. In case you use a tree, properly label the nodes in the tree.

0
1
2
3



Problem 3 (4 credits)

The World Bank reports that 1.7% of the US population lives on less than \$2 per day. A policy maker claims that this number is misleading because of variation from state to state and rural to urban. To investigate this, she takes a random sample of 100 individuals in Atlanta to compare with the national average and finds that 4% of the Atlanta population live on less than \$2 per day.

- 0 ☐ a) The policy maker wants to show that the percentage for Atlanta is significantly higher than the national average. How should she set up her null and her alternative hypothesis?

1 ☐

- 0 ☐ b) Can she draw her desired conclusion at a significance level of $\alpha = 0.05$? Justify your answer.

Hint:

```
pbinom(0:5, 100, prob = 0.017)
## [1] 0.1800329 0.4913817 0.7579127 0.9084860 0.9716334 0.9926011
```

1 ☐

2 ☐

3 ☐

Problem 4 (8 credits)

The American National Election Studies (ANES) collects data on voter attitudes and intentions as well as demographic information. In this question we will focus on two variables from the 2012 ANES dataset:

- region (levels: Northeast, North Central, South, and West),
- whether the respondent feels things in the country are generally going in the right direction or things have pretty seriously gotten off on the wrong track.

To keep calculations simple we will work with a random sample of 400 respondents from the ANES dataset. The distribution of responses is as follows:

```
addmargins(table(anes))
##           region
## direction north_central northeast south west Sum
## right_direction      38      19   46   31 134
## wrong_track          59      44  107   56 266
## Sum                 97      63  153   87 400
```

According to the 2010 Census, 18% of US residents live in the Northeast, 22% live in the North Central region, 37% live in the South, and 23% live in the West.

Given the data, the following statistical analysis was done in R.

```
chisq.test(table(anes$region), p = c(0.22, 0.18, 0.37, 0.23))
##
## Chi-squared test for given probabilities
##
## data:  table(anes$region)
## X-squared = 2.4861, df = 3, p-value = 0.4778
```

a) State the null and alternative hypothesis corresponding to the test done in the statistical analysis of the data.

0
1

b) Check the conditions of the test done in the statistical analysis.

0
1
2

c) State the conclusion of the test in the context of the data. Justify your answer.

0
1

d) Now assume we want to test the following hypotheses:

H_0 : Region and opinion on direction are independent.

H_A : Region and opinion on direction are dependent.

0
1
2

Complete the R code on the next page so that this command can be used to test the above hypotheses using the `infer` workflow.

```
library(infer)
null_dist <- anes |>

specify(formula = _____) |>

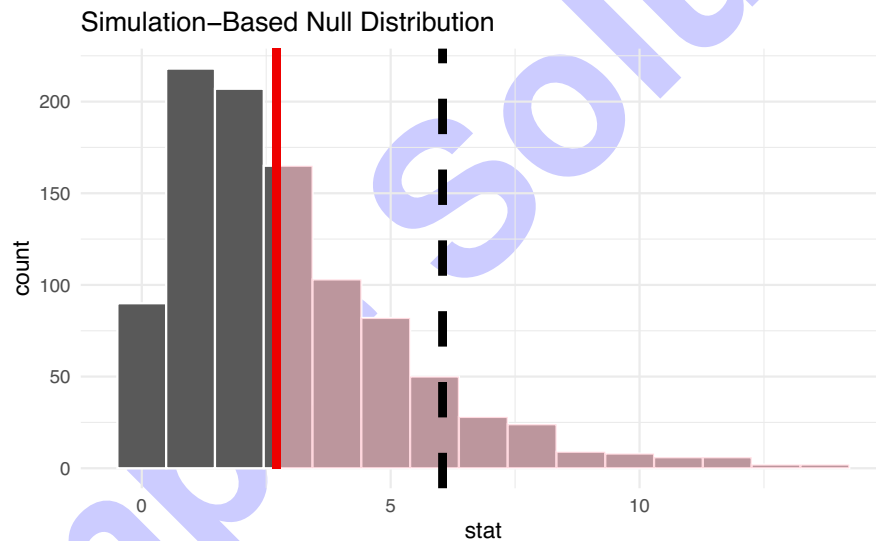
hypothesize(null = "_____") |>

_____(reps = 1000) |>

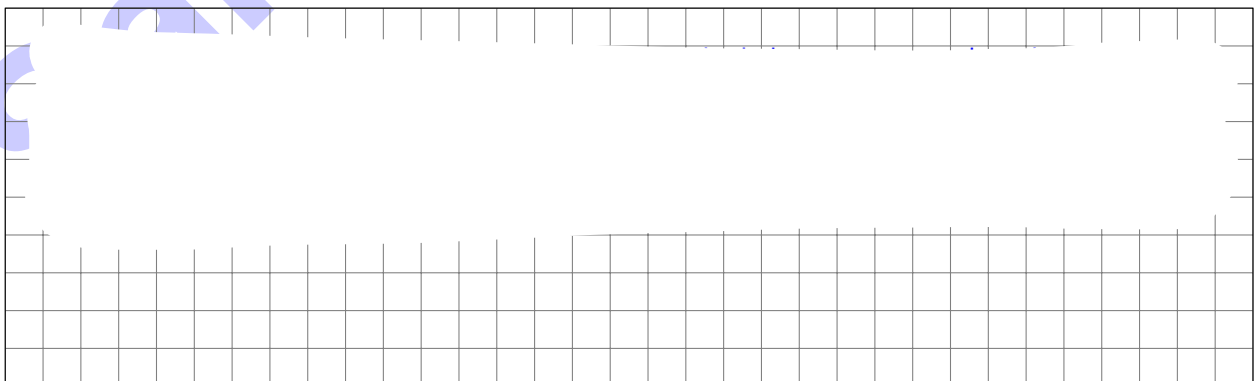
calculate(stat = "_____")
```

0 ☐ e) Applying the infer workflow leads to the following simulated null distribution and an observed value for the
1 ☐ test statistic equal to 2.7058648
2 ☐

```
qstat <- quantile(null_dist$stat, probs = 0.9)
null_dist |> infer::visualize() +
  shade_p_value(obs_stat = observed_stat, direction = "greater") +
  geom_vline(xintercept = qstat, size = 2, linetype = 2)
```



Given these values, decide if the null hypothesis can be rejected at a significance level of 0.1. Justify your answer by using a precise argument.



Problem 5 (4 credits)

For a t-test in R, we obtained the following output.

```
t = 0.2024, df = 14, p-value = 0.8425
alternative hypothesis: true mean is not equal to 20
95 percent confidence interval:
18.91483 21.31131
sample estimates:
mean of x
20.11307
```

Decide if the following statements are true or false. Justify your **answer**.

Remark: Only correct justifications are awarded with points. Not the decision (true/false).

a) The null hypothesis tested here is $H_0 : \mu = 20.11307$.

0
1

b) The following R-code `pt(0.2024, df=14)` will result in an output of 0.8425.

0
1

c) The margin of error of the computed confidence interval is equal to 1.19824.

0
1

d) [18.97138, 21.25476] is a 99% confidence interval for the true mean.

0
1

Problem 6 (3 credits)

A medical research group recruits people to complete a series of short surveys about their medical history. For example, one survey asks for information on a person's family history regarding cancer. Another survey asks questions about the person's last visit to a hospital. So far, as people sign up, the average number of surveys a person completes is only 4; the standard deviation of the number of surveys is about 2.2. The research group thus wants to try a new interface that they think will encourage new enrollees to complete more surveys. To this end, they plan an experiment in which they randomize each enrollee to either the new interface or the current interface.

0 ☐ How many new enrollees do they need for each interface to detect an effect size of 1 additional survey being completed per enrollee? To answer this question work with a significance level of 0.05, and a desired power of 70%.

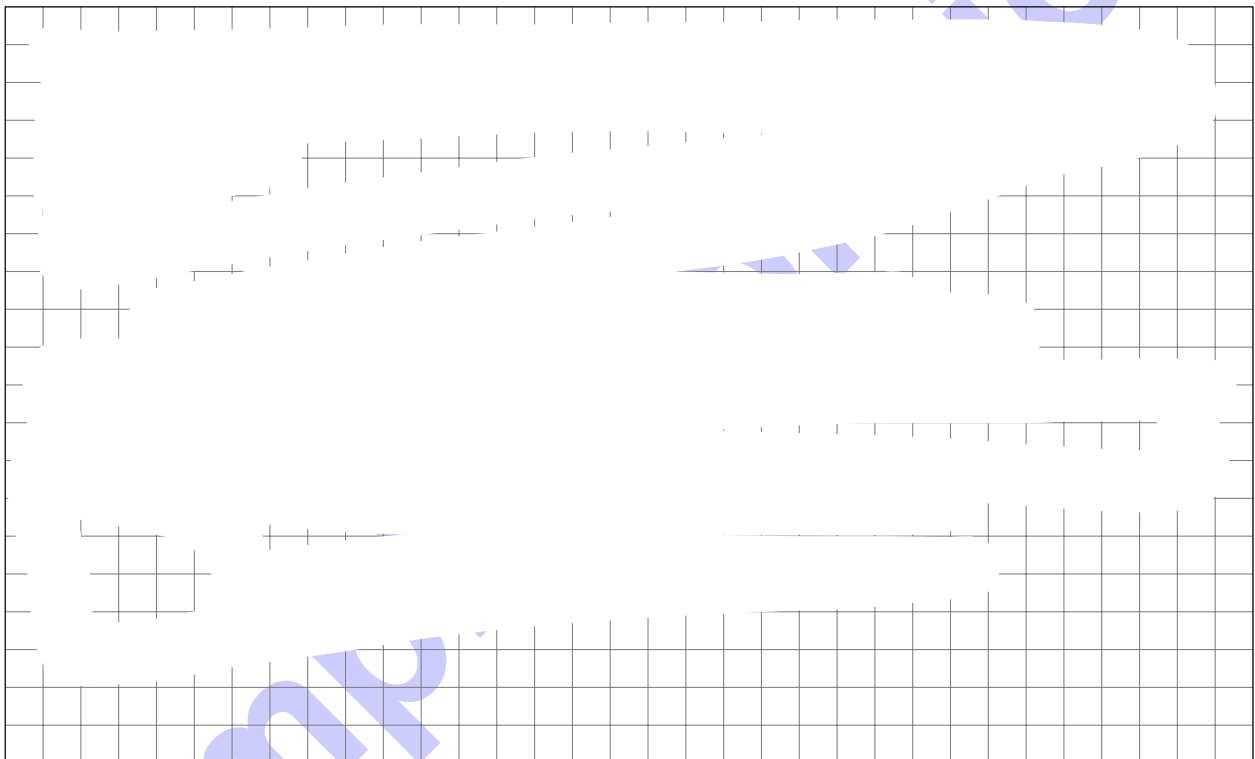
1 ☐

2 ☐

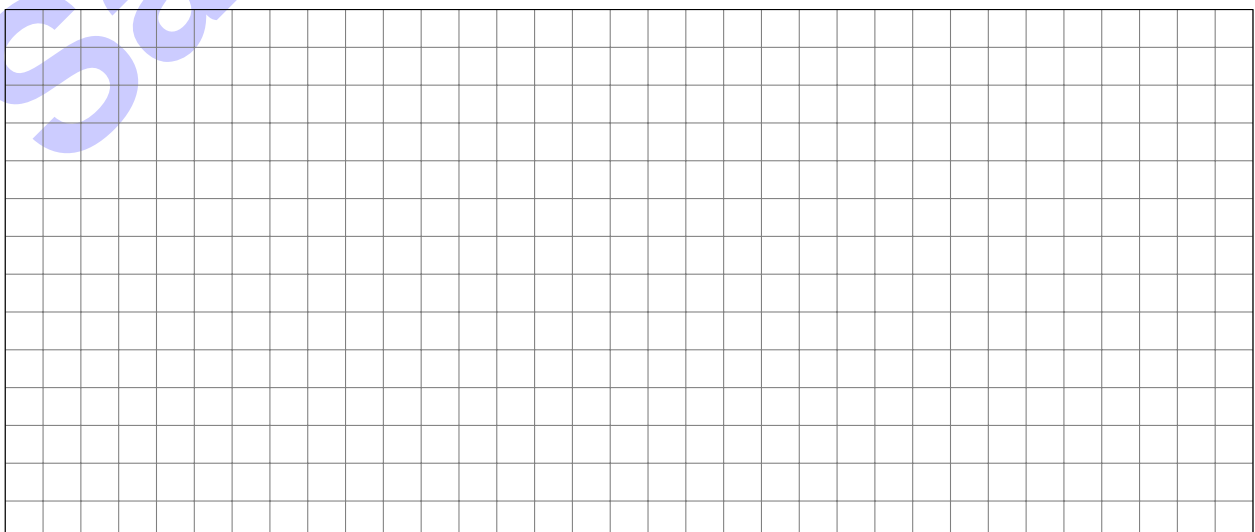
3 ☐

Hint:

```
qnorm(c(0.7, 0.8, 0.9, 0.95, 0.975, 0.99))  
## [1] 0.524 0.842 1.282 1.645 1.960 2.326
```



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.



01

01

- 0
- 1
- 2
- 3
- 4

0
1

01

- 0
- 1
- 2
- 3
- 4

[illegible]

Problem 8 (7 credits)

Spam filters are built on principles similar to those used in logistic regression, which may be applied to estimate the probability that a message is spam (or not) on the basis of characteristics of the considered message. Here, we consider a data set whose variables are either numerical or binary indicator variables. The variable names are as follows (we avoid further details for brevity):

```
names(email)
## [1] "spam"          "to_multiple"   "from"          "cc"            "sent_email"
## [6] "image"         "attach"        "dollar"         "winner"        "inherit"
## [11] "viagra"        "password"      "num_char"       "line_breaks"   "format"
## [16] "re_subj"       "exclaim_subj" "urgent_subj"    "exclaim_mess"  "number"
```

We begin by splitting the data into a training set `email_train` and a test set `email_test`. We then fit a logistic regression model to the training data, using all available predictor variables. The values of the response are 1 (for spam) and 0 (for not spam).

```
model_full <- glm(spam~., data = email_train, family = binomial)
```

- 0 ☐ a) The full model contains uninformative predictor variables, which we want to identify by running a backward selection. In one of the steps, we obtain the following values

1 ☐

```
Start:      AIC=1329.96
spam ~ to_multiple + from + cc + sent_email + image + attach +
      dollar + winner + inherit + viagra + password + num_char +
      line_breaks + format + re_subj + exclaim_subj + urgent_subj +
      exclaim_mess+number
```

	Df	Deviance	AIC
<none>		1288.0	1330.0
- viagra	1	1291.9	1331.9
- num_char	1	1292.5	1332.5
- sent_email	1	1403.1	1443.1
- inherit	1	1293.4	1333.4
- from	1	1296.3	1336.3
- format	1	1296.4	1336.4
- image	1	1299.0	1339.0
- cc	1	1288.1	1328.1
- dollar	1	1302.7	1342.7
- attach	1	1304.7	1344.7
- line_breaks	1	1305.0	1345.0
- exclaim_mess	1	1309.2	1349.2
- exclaim_subj	1	1288.2	1328.2
- urgent_subj	1	1298.1	1338.1
- re_subj	1	1310.8	1350.8
- password	1	1304.9	1344.9
- winner	1	1312.2	1352.2
- number	2	1346.2	1384.2
- to_multiple	1	1375.7	1415.7

Which variable should be removed (if any) in this step? Justify your answer.

Remark: Be aware that the output is not given in the standard form.

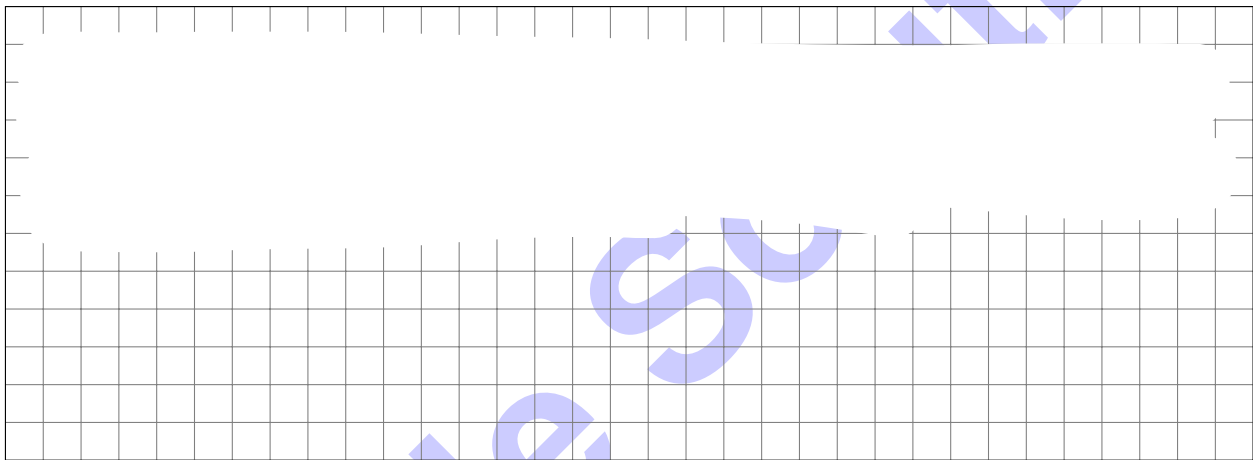
b) After some additional steps and further modifications we end up with the following model

```
tidy(red_model)
## # A tibble: 5 x 5
##   term          estimate std.error statistic  p.value
##   <chr>          <dbl>    <dbl>    <dbl>    <dbl>
## 1 (Intercept)  -1.13      0.0849    -13.3  1.56e-40
## 2 to_multiple1 -2.17      0.329     -6.62  3.60e-11
## 3 winneryes     2.02      0.369      5.48  4.31e- 8
## 4 line_breaks  -0.00436   0.000494  -8.83  1.05e-18
## 5 re_subj1     -3.03      0.387     -7.82  5.08e-15
```

containing the predictor variables: to_multiple (0=no, 1=yes), winner (yes/no term winner contained in email), the number of line_breaks and re_subj (1= subject started with "Re:", 0= subject didn't started with "Re:").

Give an **interpretation** of the estimated slopes of line_breaks and re_subj in terms of the odds ratio.

Remark: Computing the odds ratio is necessary, but not sufficient. You should describe in words the meaning of the computed value.



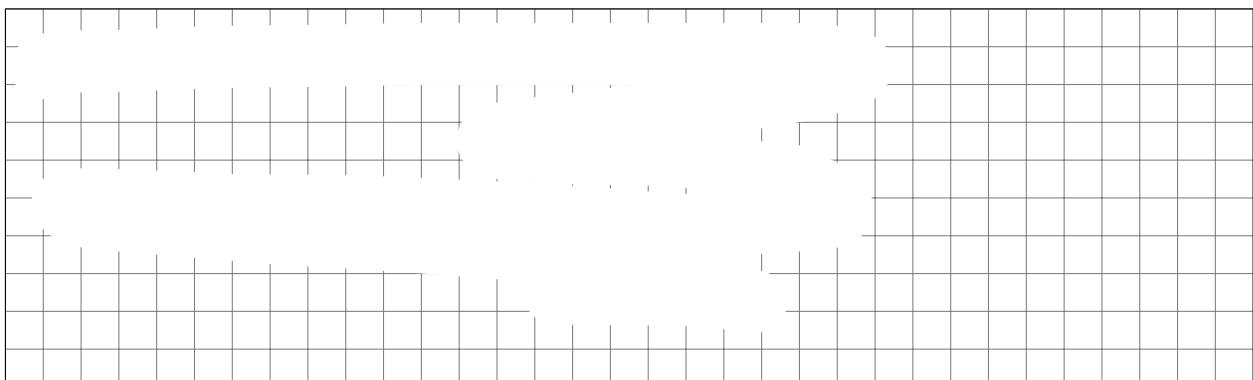
c) Given the full model, we are now able to compute predictions for the test dataset.

```
email_test$pred_prob <- predict(model_full, newdata = email_test, type = "response")
```

To classify an email as spam or not, we need to threshold the estimated probabilities of being spam. Choosing a threshold of 0.5 leads to the following table

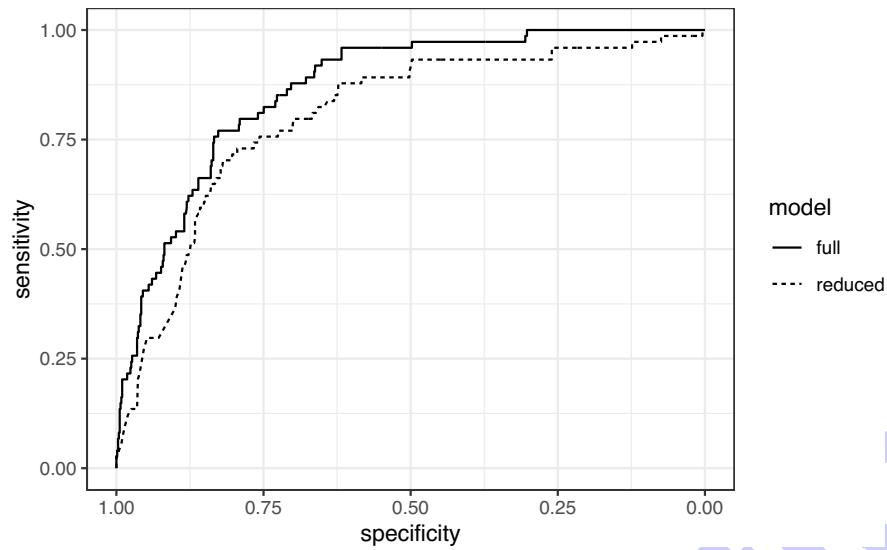
```
##           Truth
## Prediction FALSE TRUE
##      FALSE   706   64
##      TRUE     5   10
```

Compute sensitivity and specificity of the predictions.

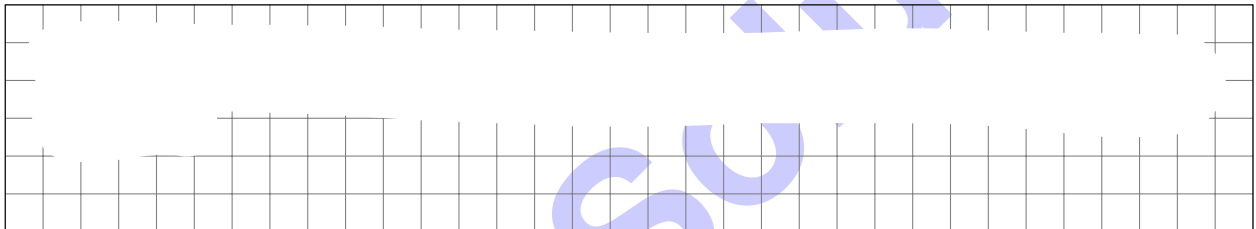


- 0 ☐
- 1 ☐
- 2 ☐

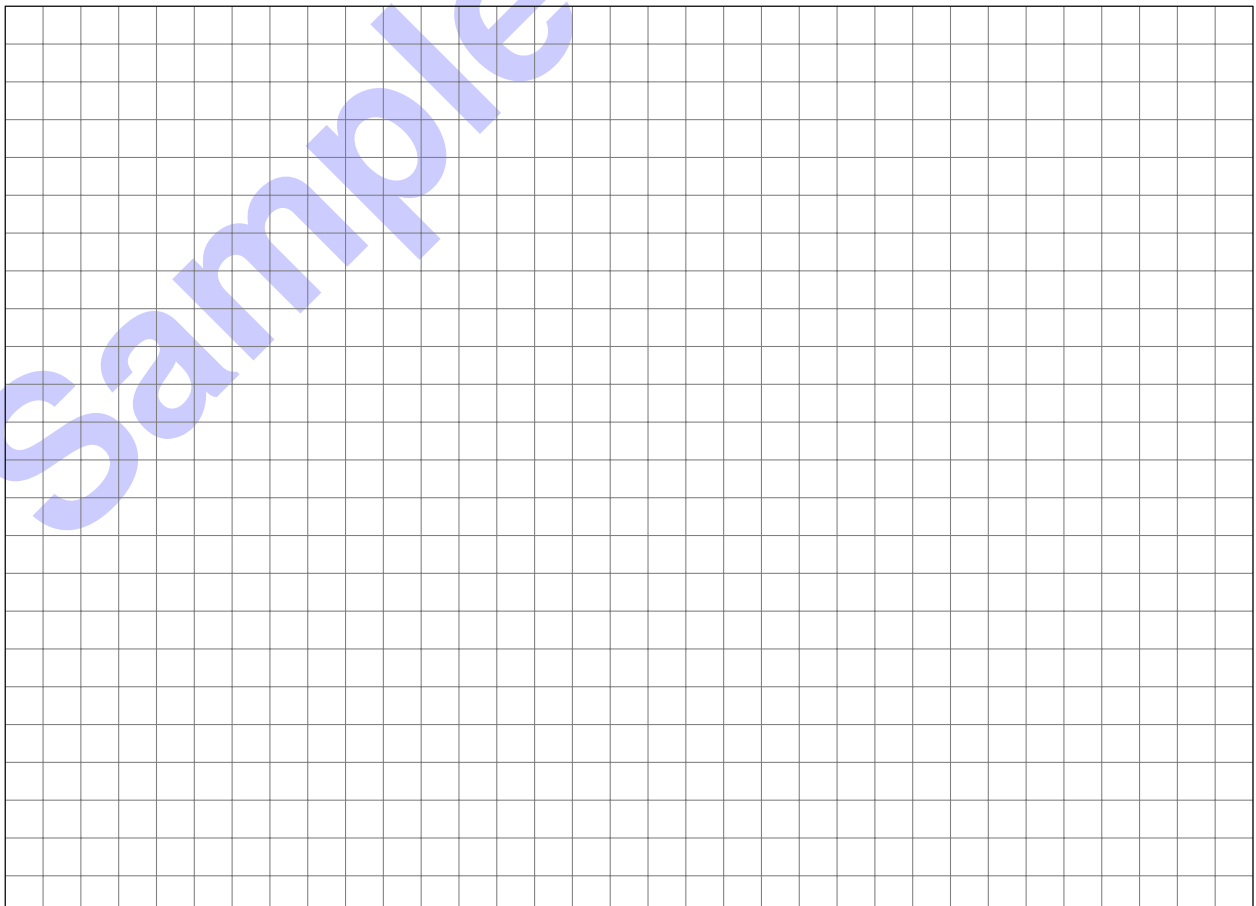
d) The following graphic contains the ROC curves of the full and reduced model.



The AUC value of the full model is equal to 0.867. Decide if the reduced model has a smaller or larger AUC value. Justify your answer.



Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.



Cheat sheet

Exploring data:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad s = \sqrt{s^2} \quad IQR = Q_3 - Q_1 \quad Q_1 - 1.5 \cdot IQR, \quad Q_3 + 1.5 \cdot IQR$$

$$r_{(x,y),n} = \frac{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{s_x \cdot s_y}$$

Probability:

$$P(A^c) = 1 - P(A); \quad P(A \cap B) = P(A) \cdot P(B), \quad A, B \text{ independent}; \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}; \quad E(X) = \sum_{i=1}^k x_i P(X = x_i), \quad E(X) = \int_{\mathbb{R}} x \cdot f(x) dx; \quad \text{Var}(X) = E(X^2) - E(X)^2;$$

$$SD(X) = \sqrt{\text{Var}(X)}; \quad \text{Cov}(X, Y) = E(X \cdot Y) - E(X) \cdot E(Y); \quad Z = \frac{X - E(X)}{\sqrt{\text{Var}(X)}}$$

$$\text{Geom. dist.: } P(X = x) = (1 - p)^{x-1} p, \quad x \in \mathbb{N}; \quad E(X) = \frac{1}{p}, \quad SD(X) = \sqrt{\frac{1-p}{p^2}}$$

$$\text{Binom. dist.: } P(X = x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}, \quad x \in \{0, 1, 2, \dots, n\}; \quad E(X) = np, \quad SD(X) = \sqrt{np(1-p)}$$

$$\text{Neg. Binom. dist.: } P(X = x) = \frac{(x-1)!}{(k-1)!(x-k)!} p^k (1-p)^{x-k}, \quad x \geq k$$

$$\text{Pois. dist.: } P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x \in \mathbb{N}_0; \quad E(X) = \lambda, \quad SD(X) = \sqrt{\lambda}$$

Inference for categorical data:

$$\hat{p} \pm z^* \cdot SE(\hat{p}) \quad SE(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \quad Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \cdot SE(\hat{p}_1 - \hat{p}_2) \quad SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

$$Z = \frac{(\hat{p}_1 - \hat{p}_2) - 0}{\sqrt{\hat{p}_{\text{pool}}(1-\hat{p}_{\text{pool}}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} \quad \hat{p}_{\text{pool}} = \frac{\hat{p}_1 n_1 + \hat{p}_2 n_2}{n_1 + n_2}$$

$$\chi^2 = \sum \frac{(\text{observed} - \text{expected})^2}{\text{expected}}, \quad \text{expected}_{ij} = \frac{\text{row } i \text{ total} \cdot \text{column } j \text{ total}}{\text{table total}} \text{ (ind.)}, \quad df = k - 1 \text{ or } (r - 1) \cdot (c - 1)$$

Inference for numerical data:

$$\bar{x} \pm t_{df}^* \cdot SE(\bar{x}) \quad T_{df}(\mathbf{x}) = \frac{\bar{x} - \mu_0}{SE(\bar{x})} \quad SE(\bar{x}) = \frac{s(x)}{\sqrt{n}} \quad df = n - 1$$

$$(\bar{x} - \bar{y}) \pm t_{df}^* \cdot SE(\bar{x} - \bar{y}) \quad T_{df}(\mathbf{x}, \mathbf{y}) = \frac{(\bar{x} - \bar{y}) - 0}{SE(\bar{x} - \bar{y})} \quad SE(\bar{x} - \bar{y}) = \sqrt{\frac{s(x)^2}{n_1} + \frac{s(y)^2}{n_2}} \quad df = \min(n_1 - 1, n_2 - 1)$$

$$\text{two-samples: } n \geq \frac{2 \cdot \sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2; \quad \text{one-sample: } n \geq \frac{\sigma_0^2}{\delta_0^2} (z_{1-\alpha^*} + z_{1-\beta})^2 \text{ with } \alpha^* \in \{\alpha, \alpha/2\}$$

ANOVA and linear regression:

$$F = \frac{MSG}{MSE} \quad df_1 = k - 1, \quad df_2 = n - k, \text{ for } k \text{ groups}; \quad SSG(\mathbf{y}) = \sum_{i=1}^k n_i (\bar{y}_i - \bar{y})^2, \quad SST(\mathbf{y}) = \sum_{j=1}^n (y_j - \bar{y})^2,$$

$$SSE(\mathbf{y}) = SST(\mathbf{y}) - SSG(\mathbf{y}); \quad p_{adj,i} = K \cdot p_i$$

$$\hat{y} = b_0 + b_1 x \quad b_1 = r_{(x,y),n} \cdot \frac{s_y}{s_x} \quad \bar{y} = b_0 + b_1 \bar{x} \quad \text{residual} = y - \hat{y} \quad \hat{\sigma} = \sqrt{\frac{1}{n-2} \sum (y_i - \hat{y}_i)^2}$$

$$T_{df}(\mathbf{y}, \mathbf{x}) = \frac{b_1(\mathbf{y}, \mathbf{x}) - \beta_{1,0}}{SE_{b_1}(\mathbf{y}, \mathbf{x})} \quad b_1(\mathbf{y}, \mathbf{x}) \pm t_{df}^* \cdot SE_{b_1}(\mathbf{y}, \mathbf{x}) \quad df = n - 2; \quad R^2 = \frac{\sum_{i=1}^n (\hat{y}_i - \bar{y})^2}{\sum_{i=1}^n (y_i - \bar{y})^2} = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

$$R_{adj}^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2 / (n - k - 1)}{\sum_{i=1}^n (y_i - \bar{y})^2 / (n - 1)}; \quad AIC = \text{constant} + n \cdot \ln \left(\sum_{i=1}^n e_i^2 \right) + 2k$$

Logistic regression:

$$\text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 x_{1,i} + \dots + \beta_k x_{k,i} \quad OR_j = \frac{\text{odds when } x_{1,j}=x+1}{\text{odds when } x_{1,j}=x} = e^{\beta_j}$$

$$\text{sensitivity} = P(\text{Test} + | \text{Condition} +), \quad \text{specificity} = P(\text{Test} - | \text{Condition} -)$$

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

A large grid of graph paper for solutions, with a diagonal watermark reading "Sample Solution".